

Definition and first terms

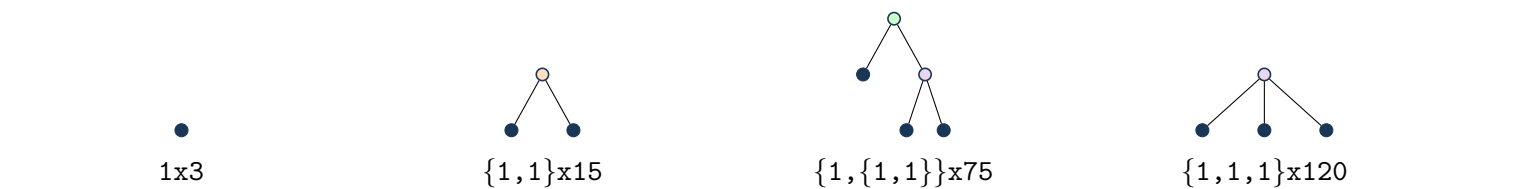
37 $A(x) = 36 + 81\,x + A(x)^{10}$, with the origin-normalized solution.

Normalization/grammar: $A(x)=1+(3)*T(x)$; T-polynomial coefficients are the Delta multiplicities below.

$a(0),\dots,a(7) = 1, 3, 15, 270, 5505, 124818, 3028200, 76896180$.

Kernel used throughout: $\rho(u) = u\,(1 - 5\,u^1 - 40\,u^2 - 210\,u^3 - 756\,u^4 - 1890\,u^5 - 3240\,u^6 - 3645\,u^7 - 2430\,u^8 - 729\,u^9)$, so $x = \rho(T(x))$.

Typogeometry



colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_{10} : 729, Δ_2 : 5, Δ_3 : 40, Δ_4 : 210, Δ_5 : 756, Δ_6 : 1890, Δ_7 : 3240, Δ_8 : 3645, Δ_9 : 2430.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_9) \in \mathbb{Z}_{\geq 0}^9 : \sum_{j=1}^9 j k_j = n - 1\}, \quad K = \sum_{j=1}^9 k_j,$$

$$a(n) = \frac{3}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_9} 5^{k_1} 40^{k_2} 210^{k_3} 756^{k_4} 1890^{k_5} 3240^{k_6} 3645^{k_7} 2430^{k_8} 729^{k_9}.$$

$$a(n) = \frac{3}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^9 P_r(n)a(n+r) = 0$ for $n \geq 1$. The exact degree-9 coefficient polynomials are embedded in `certificate_payload.json` (source: `release/certificate_payload.json.gz#objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 9. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$R(n,u) = N(n,u)/\rho(u)^8, \; \deg_u N = 81.$$

$$\text{Telescoping identity: } \sum_r P_r(n)H_{n+r}(u) = \partial_u(R(n,u)H_n(u)).$$

Matrix dimensions: 20×20 ; remainder 9×10 , rank 9, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.